

DATA MINING NOTES

UNIT – 1

DEFINITION OF MINING:

In general terms, “Mining” is the process of extraction of some valuable material from the earth. e.g. coal mining, diamond mining etc.

- In the context of computer science, “Data Mining” refers to the extraction of useful information from a bulk of data or [data warehouses](#).
- One can see that the term itself is a little bit confusing.
- In case of coal or diamond mining, the result of extraction process is coal or diamond.
- But in case of Data Mining, the result of extraction process is not data!! Instead, the result of data mining is the patterns and knowledge that we gain at the end of the extraction process.
- In that sense, Data Mining is also known as Knowledge Discovery or Knowledge Extraction.

WHAT IS DATA MINING?

- Data mining is the process of extraction of interesting (nontrivial, implicit, previously unknown, comprehensive) patterns or knowledge from huge amount of data. It is the set of activities used to find new, hidden or unexpected patterns in data or unusual patterns in data.

- Data mining is one of the fastest growing fields in the computer industry. At present it is the most popular research area in computer science
- Data mining is the process of analyzing hidden patterns of data according to different perspectives for categorization into useful information, which is collected and assembled in common areas, such as data warehouses, for efficient analysis, data mining algorithms, facilitating business decision making and other information requirements to ultimately cut costs and increase revenue.
- Data mining is also known as data discovery and knowledge discovery.
- **Data:** Data are any facts, numbers or text that can be processed by a computer.
- Today, organizations are accumulating vast and growing amounts of data in different formats and different databases includes.
- **Operational or transactional** :data such as sales, cost, inventory payroll and accounting
- **No Operational data** such as industry sales, forecast data, macro economic data.
- **Meta data** – data about the data itself, such as logical data base design or data dictionary definitions.
- **Information:** The patterns, associations or relationships among all this data can provide information. For example analysis of

retail point of sale transaction data can yield information on which products are selling and when.

- **Knowledge:** knowledge is integrated information, including facts and their relation, which have been perceived, discovered, or learned as our “mental pictures”
- Information can be converted into knowledge about historical patterns and future trends.
- Summary information on retail supermarket sales can be analyzed in light of promotional efforts to provide knowledge of consumer buying behavior. Thus , a manufacturer or retailer could determine which items are most susceptible to promotional efforts.

Historical Note: Many names of Data Mining

- Data Mining and data knowledge discovery field has been called by many names
- In 1960-s statisticians have used terms like “Data fishing” or “Data Dredging”
- The term “Data Mining” appeared around 1990 in the database community. Briefly there was a phrase “data base mining”TM (trade mark by HNC) and researchers turned to “data mining”
- Other terms used include data archaeology, Information Harvesting, Information discovery, Knowledge Extraction , etc.,
- George Piatetsky- Shapiro coined the term “knowledge Discovery in Databases”- 1989.

- The term data mining become more popular in business community and in the press.
- As of Jan 2004, Google search for “data mining” finds over 20,00,000 pages.
- Search for “Knowledge discovery” finds only 3,00,000 pages.
- In 2003 – “data mining” has acquired bad image because of its association with US government program of TIA (Total Information Awareness).
- Headlines such as “senate kills data mining program”, Computer world, July 18, 2003, referring to US Senate decision to close down TIA.
- Currently, Data Mining and Knowledge Discovery are used interchangeably, and we also use these terms synonyms.

The Evolution of Data Mining

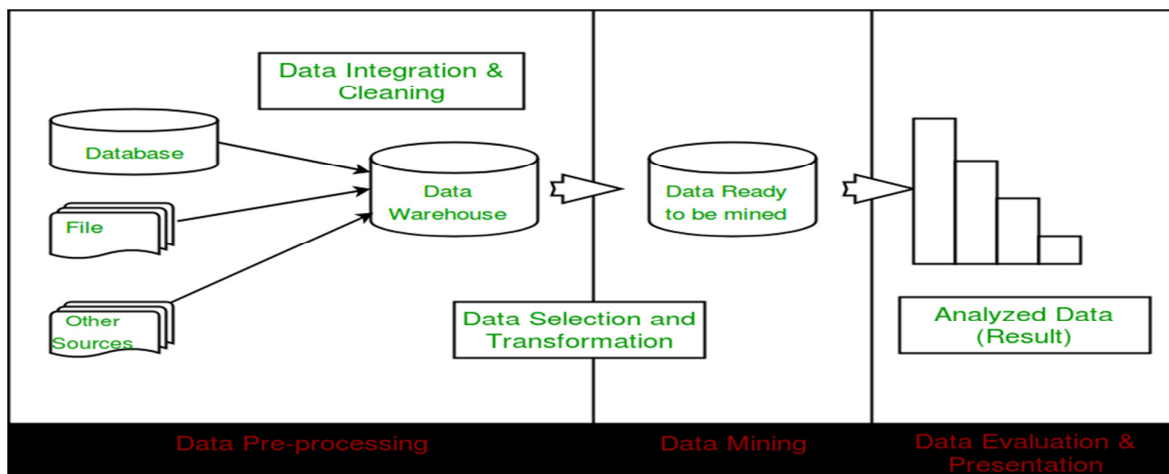
Data mining is a natural development of the increased use of computerized databases to store data and provide answers to business analysts.

Evolutionary Step	Business Question	Enabling Technology
Data Collection (1960s)	"What was my total revenue in the last five years?"	computers, tapes, disks
Data Access (1980s)	"What were unit sales in New England last March?"	faster and cheaper computers with more storage, relational databases
Data Warehousing and Decision Support	"What were unit sales in New England last March? Drill down to Boston."	faster and cheaper computers with more storage, On-line analytical processing (OLAP), multidimensional databases , data warehouses
Data Mining	"What's likely to happen to Boston unit sales next month? Why?"	faster and cheaper computers with more storage, advanced computer algorithms

Data Mining as a whole process

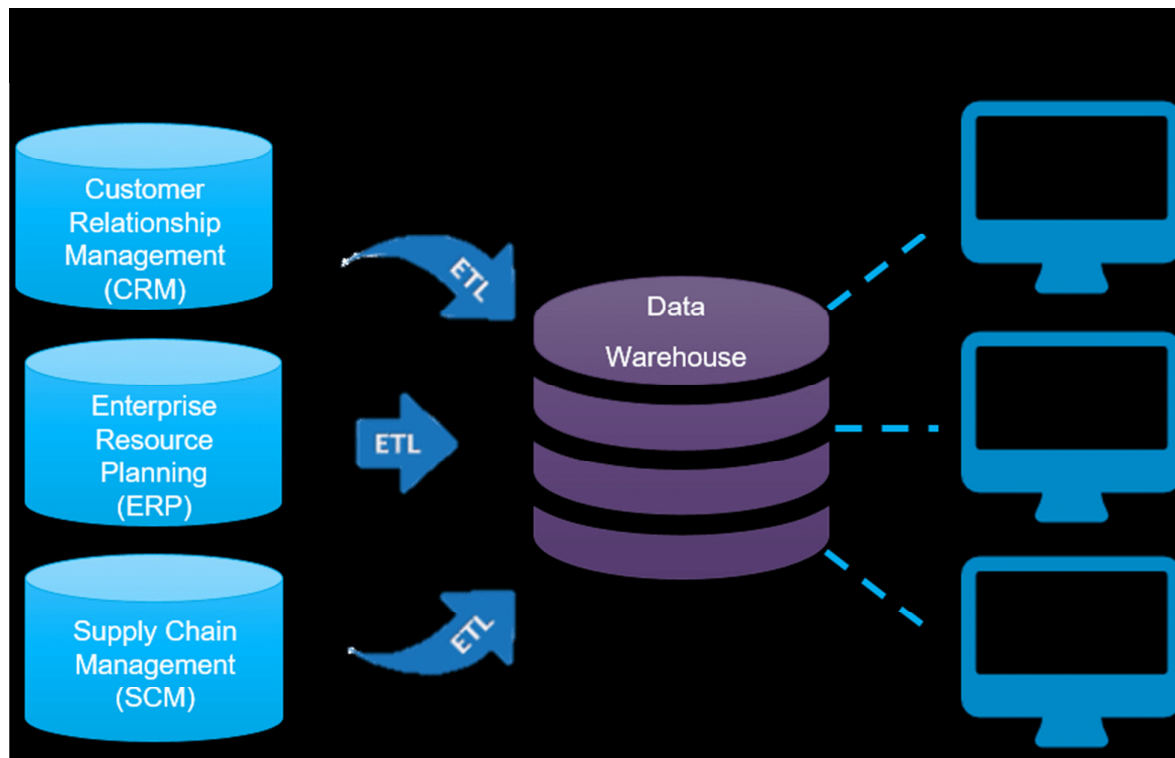
The whole process of Data Mining comprises of three main phases:

1. Data Pre-processing – Data cleaning, integration, selection and transformation takes place
2. Data Extraction – Occurrence of exact data mining
3. Data Evaluation and Presentation – Analyzing and presenting results



DATA WAREHOUSE

- A data warehouse is a [repository](#) for data generated and collected by an enterprise's various operational systems.
- Data warehousing is often part of a broader [data management strategy](#) and emphasizes the capture of data from different sources for access and analysis by business analysts, data scientists and other end users.



What is data warehousing?

- A data warehouse is a database designed to support decision making in an organization.
- Data from the production databases are copied to the data warehouse so that queries can be performed without disturbing the performance or the stability of the production systems.
- For data mining to occur, it is crucial that data warehousing is present.

An example of how well data warehousing and data mining has been utilized is Walmart. Walmart maintains a 7.5 TB data warehouse. Retailers capture Point of Sale (POS) transaction data from over 2,900 stores across 6 countries and transmit them to Walmart's

data warehouse. Walmart then allows their suppliers to access the data to collect information

- on their products to analyse how they can improve their sales.
- These suppliers will then better understand customer buying patterns and manage local store inventory, etc
- Data Mining Functionalities
- This huge amount of data must be processed in order to extract useful information and knowledge since they are not explicit. Data Mining is the process of discovering interesting knowledge from a large amount of data.

The kinds of patterns that can be discovered depend upon the data mining tasks employed. By and large, there are two types of data mining tasks: descriptive data mining tasks that describe the general properties of the existing data, and predictive data mining tasks that attempt to do predictions based on inference on available data.

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Descriptive mining:

- This term is basically used to produce correlation, cross-tabulation, frequency etc. These technologies are used to determine the similarities in the data and to find existing patterns. One more application of descriptive analysis is to develop the captivating subgroups in the major part of the data available.

This analytics emphasis on the summarization and transformation of the data into meaningful information for reporting and monitoring.

Predictive Data Mining:

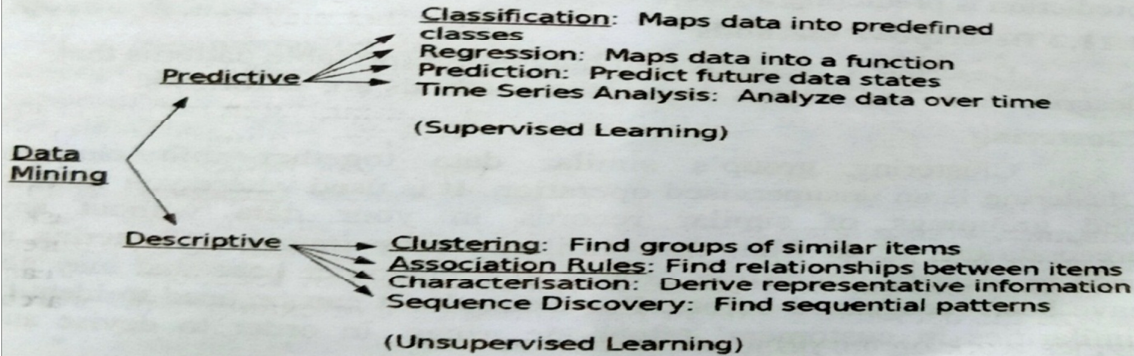
The main goal of this mining is to say something about future results not of current behaviour. It uses the supervised learning functions which are used to predict the target value. The methods come under this type of mining category are called classification, time-series analysis and regression.

- Modelling of data is the necessity of the predictive analysis, and it works by utilizing a few variables of the present to predict the future not known data values for other variables.

1.11 Basic Data Mining Tasks / Taxonomy of data mining tasks

The various basic data mining tasks are as follows:

Predictive	Description
Classification	Clustering
Regression	Summarization
Time series analysis	Association Rules
Prediction	Sequence discovery



1.10

1.11.1 Prediction methods

It uses some variables to predict unknown or future values of the same or other variables. The various types of prediction methods are as follows:

Classification

Classification maps the data into predefined group or classes. Stored data is used to locate data in predetermined groups. It is often referred to as supervised learning because the classes are determined before examining the data.

Regression

Regression is used to map a data item to a real valued predictions variable. Regression involves the learning of the function that does this mapping. Regression assumes that the target data fit into some known type of function and determine the best function of this type that models the given data.

Time series analysis

The value of an attribute is examined as it varies over time. The values are usually are obtained as evenly spaced time points (daily, weekly, hourly etc.)

Prediction

Prediction can be viewed as a type of classification. This is a data mining task that is different from the prediction model, although the prediction task is a type of prediction model. The difference is that prediction is predicting a future state rather than a current state.

1.11.2 Descriptive methods

Descriptive methods find human interpretable patterns that describe data. The various descriptive methods are as follows:

Clustering

Clustering group's similar data together into clusters. Clustering is an unsupervised operation. It is used where you wish to find groupings of similar records in your data without any preconditions as to what that similarity may involve. Clustering is used to identify interesting groups in a customer base that may not have been recognized before. For example, it can be used to identify similarities in customers' telephone usage, in order to devise and market new call services.

Association Rules

It is a model that identifies specific types of data associations. These associations are often used in the retail sales community to identify items that are frequently purchased together. Data can be mined to identify associations.

Sequence Pattern Discovery

Sequential analysis or sequence discovery is used to determine sequential patterns in data. In other words, we can say that data is mined to anticipate behavior patterns and trends. These

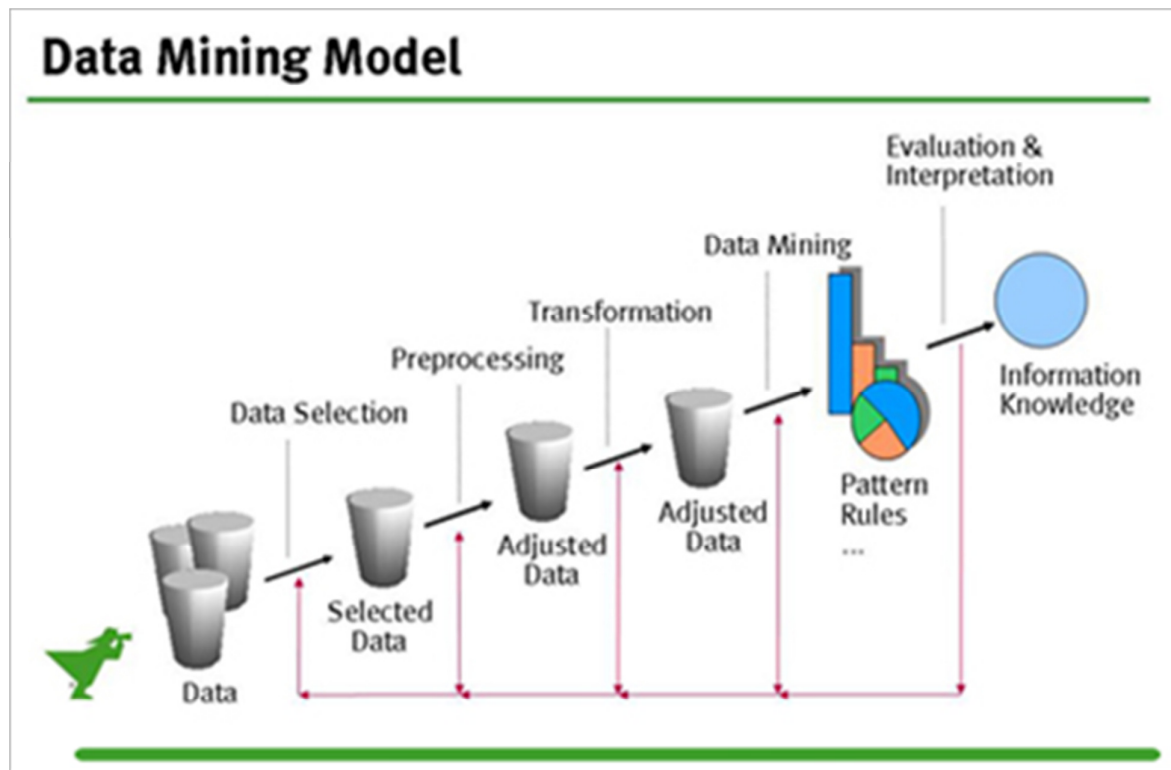
Characteristics	Data Mining	Data Base
Query	Poorly Defined	Well defined
	No precisely query language	SQL (Structured Query Language)
Data	Non operational data	Operational data
Output	Fuzzy	Precise
	Not a subset of database	Subset of database

patterns are based on a time sequence of actions. These patterns are similar to associations in that data are found to be related, but the relationship is based on time.

Summarization

Summarization maps data into subsets with associated simple descriptions. Summarization is also called characterization or generalization. It extracts or derives representative information about the database. This may be accomplished by actually retrieving portions of data.

STEPS IN DATA MINING PROCESS



THE KNOWLEDGE DISCOVERY PROCESS IS A SEQUENCE OF THE FOLLOWING STEPS:

1. **Data Cleaning:** This step removes noise and inconsistent data from the input data.
2. **Data Integration:** This step combines multiple sources of data. The data cleaning and data integration step together to form the preprocessing of data. The preprocessed data is then stored in the data warehouse.
3. **Data Selection:** These steps select the data to the analysis task from the database.

4. **Data Transformation:** In this step, various data aggregation and data summary techniques are applied to transform the data into a useful form for mining.

5. **Data Mining:** In this step, data patterns are extracted by applying intelligent methods.

6. **Pattern Evaluation:** The extracted data patterns are evaluated and recognized according to the interestingness measures.

7. **Knowledge Representation:** Visualization and knowledge representation techniques are used to present the mined knowledge to the users.

MAJOR ISSUES IN DATA MINING / DATA MINING ISSUES

Data mining systems depend on databases to supply the raw input and this raises problems, such as that database tends to be dynamic, incomplete, noisy and large. The major issues in data mining systems are as follows:

- Mining methodology and user interaction
- Performance Issues
- Issues relating to the diversity of data types

Mining methodology and user interaction

□ **Mining different kinds of knowledge in databases** – Different users can be interested in different kinds of knowledge. Data mining should cover a wide spectrum of data analysis and knowledge discovery tasks which may use the same database in different ways.

Interactive mining of knowledge at multiple levels of abstraction
– allows users to focus the search for patterns, providing and refining

data mining requests based on returned results to view data and discovered patterns at multiple granularities.

Incorporation of background knowledge – Background knowledge guides the discovery process in concise terms at different levels of abstraction.

Data mining query languages and ad-hoc data mining – Data mining query languages need to be developed to allow users to describe ad hoc data mining tasks by facilitating the specification of the relevant data.

Expression and visualization of data mining results – This requires the system to adopt expressive knowledge representation techniques, such as graph, trees, tables, rules, charts, matrices or curves.

Handling noise and incomplete data - Noise or data may confuse the process, causing the knowledge model constructed to overfit the data.

Pattern evaluation: the interesting nature of the problem – Data mining system can uncover thousands of patterns. Many of the patterns discovered may be uninteresting to the given user.

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 data.

Performance Issues

Efficiency and scalability of data mining algorithms – Extract information from a large amount of data in databases.

Parallel, distributed and incremental mining methods – The large size of databases, wide distribution of data, high cost and the computational complexity of data mining methods lead to the development of parallel and distributed data mining algorithm.

Issues relating to the diversity of data types

Handling relational and complex types of data – Specific data mining systems should be constructed for mining specific kinds of data. Therefore, one may expect to have different data mining systems for different kinds of data.

□ *Mining information from heterogeneous databases and global information systems (WWW)* – Data mining may help data regularities in multiple heterogeneous databases that are unlikely to be discovered by simple query systems and may improve information exchange and interoperability in heterogeneous databases.

DATA MINING - APPLICATIONS

Financial Data Analysis

The financial data in banking and financial industry is generally reliable and of high quality which facilitates systematic data analysis and data mining. Some of the typical cases are as follows –

Design and construction of data warehouses for multidimensional data analysis and data mining.

Loan payment prediction and customer credit policy analysis.

Classification and clustering of customers for targeted marketing.

Detection of money laundering and other financial crimes.

Retail Industry

Data Mining has its great application in Retail Industry because it collects large amount of data from on sales, customer purchasing history, goods transportation, consumption and services. It is natural that the quantity of data collected will continue to expand rapidly because of the increasing ease, availability and popularity of the web.

Data mining in retail industry helps in identifying customer buying patterns and trends that lead to improved quality of customer service and good customer retention and satisfaction. Here is the list of examples of data mining in the retail industry –

Design and Construction of data warehouses based on the benefits of data mining.

Multidimensional analysis of sales, customers, products, time and region.

Analysis of effectiveness of sales campaigns.

Customer Retention.

Product recommendation and cross-referencing of items.

EXAMPLES OF DATA MINING IN REAL LIFE

The importance of data mining and analysis is growing day by day in our real life. Today most organizations use data mining for analysis of Big Data.

Let us see how these technologies benefit us.

Mobile Service Providers

Mobile service providers use data mining to design their marketing campaigns and to retain customers from moving to other vendors.

From a large amount of data such as billing information, email, text messages, web data transmissions, and customer service, the data mining tools can predict “churn” that tells the customers who are looking to change the vendors.

With these results, a probability score is given. The mobile service providers are then able to provide incentives, offers to customers who are at higher risk of churning. This kind of mining is often used by major service providers such as broadband, phone, gas providers, etc.

Retail Sector

Data Mining helps the supermarket and retail sector owners to know the choices of the customers. Looking at the purchase history of the customers, the data mining tools show the buying preferences of the customers.

With the help of these results, the supermarkets design the placements of products on shelves and bring out offers on items such as coupons on matching products, and special discounts on some products.

These campaigns are based on RFM grouping. RFM stands for recency, frequency, and monetary grouping. The promotions and marketing campaigns are customized for these segments. The customer who spends a lot but very less frequently will be treated differently from the customer who buys every 2-3 days but of less amount.

Data Mining can be used for product recommendation and cross-referencing of items.

Artificial Intelligence

A system is made artificially intelligent by feeding it with relevant patterns. These patterns come from data mining outputs. The outputs of the artificially intelligent systems are also analyzed for their relevance using the data mining techniques.

The recommender systems use data mining techniques to make personalized recommendations when the customer is interacting with the machines. The artificial intelligence is used on mined data such as giving product recommendations based on the past purchasing history of the customer in Amazon.

Ecommerce

Many E-commerce sites use data mining to offer cross-selling and upselling of their products. The shopping sites such as Amazon, Flipkart show “People also viewed”, “Frequently bought together” to the customers who are interacting with the site.

These recommendations are provided using data mining over the purchasing history of the customers of the website.

Science And Engineering

With the advent of data mining, scientific applications are now moving from statistical techniques to using “collect and store data” techniques, and then perform mining on new data, output new results and experiment with the process. A large amount of data is collected

from scientific domains such as astronomy, geology, satellite sensors, global positioning system, etc.

Data mining in computer science helps to monitor system status, improve its performance, find out software bugs, discover plagiarism and find out faults. Data mining also helps in analyzing the user feedback regarding products, articles to deduce opinions and sentiments of the views.

Crime Prevention

Data Mining detects outliers across a vast amount of data. The criminal data includes all details of the crime that has happened. Data Mining will study the patterns and trends and predict future events with better accuracy.

The agencies can find out which area is more prone to crime, how much police personnel should be deployed, which age group should be targeted, vehicle numbers to be scrutinized, etc.

Research

Researchers use Data Mining tools to explore the associations between the parameters under research such as environmental conditions like air pollution and the spread of diseases like asthma among people in targeted regions.

Farming

Farmers use Data Mining to find out the yield of vegetables with the amount of water required by the plants.

Dynamic Pricing

Data mining helps the service providers such as cab services to dynamically charge the customers based on the demand and supply. It is one of the key factors for the success of companies.

Transportation

Data Mining helps in scheduling the moving of vehicles from warehouses to outlets and analyze the product loading patterns.

Insurance

Data mining methods help in forecasting the customers who buy the policies, analyze the medical claims that are used together, find out fraudulent behaviors and risky customers.

UNIT -2

Data Preprocessing:

Data preprocessing is a data mining technique that involves transforming raw data into an understandable format. Real-world data is often incomplete, inconsistent, and/or lacking in certain behaviors or trends, and is likely to contain many errors. Data preprocessing is a proven method of resolving such issues. Data preprocessing prepares raw data for further processing.

mining purpose. Data is captured and prepared for data

6.2 Why preprocess the data? / Need for preprocessing

We need *clean* data to produce good results

- Data in the real world is dirty
 - *Incomplete*: lacking attribute values, lacking certain attributes of interest, or containing only aggregate data
 - *Noisy*: containing errors or outliers
 - *Inconsistent*: containing discrepancies in codes or names
- No quality data, no quality mining results!
 - Quality decisions must be based on quality data
 - Data warehouse needs consistent integration of quality data

6.3 Data Preprocessing Techniques / Major Tasks in Data Preprocessing

The major tasks in data preprocessing are as follows:

- **Data Cleaning**
 - Fill in missing values, smooth noisy data, identify or remove outliers, and resolve inconsistencies
- **Data Integration**
 - Integration of multiple databases, data cubes, or files
- **Data Transformation**
 - Normalization and aggregation.

- **Data Reduction**
 - Obtains reduced representation in volume but produces the same or similar analytical results.
- **Data Discretization**
 - Part of data reduction but with particular importance, especially for numerical data.

Summarization of the tasks of pre processing.

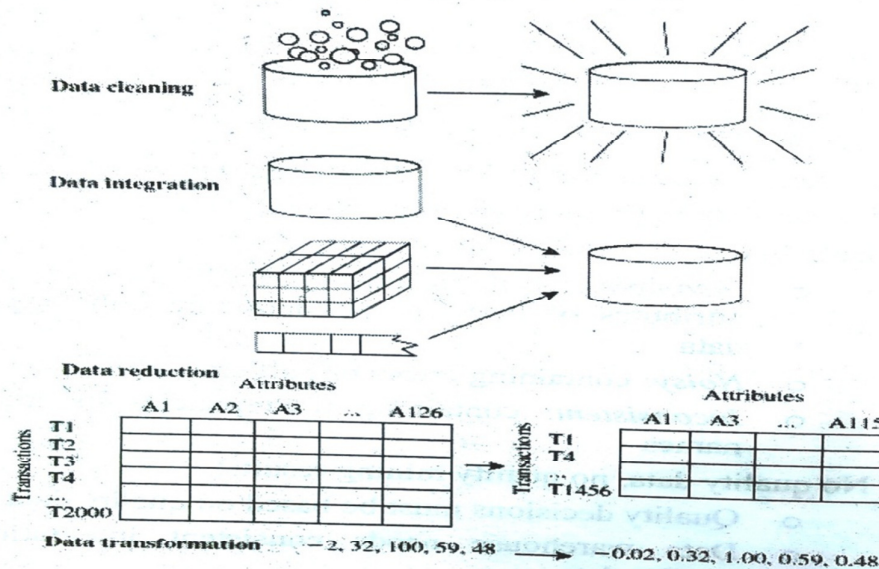


Figure: Forms of data preprocessing.

6.4 Data Cleaning

Data cleaning tasks are used to fill in missing values, identifying outliers and smooth out noisy data, correct inconsistent data and smooth the noisy data.

6.4.1 Missing Data / Values

Data is not always available. For example, many tuples have no recorded value for several attributes, such as customer income in sales data.

Missing data may be due to (i) equipment malfunction,

(ii) inconsistent with other recorded data and thus deleted, (iii) data not entered due to misunderstanding, as certain data may not be considered important at the time of entry. Missing data needs to be inferred.

6.4.1.1 Methods of handling missing data

- **Ignore the tuple:** This is usually done when class label is missing. This method is not very effective, unless the tuple contains several attributes with missing values. *Return*
- **Fill in the missing value manually:** Manually search for all missing values and replace them with appropriate values. In general, this approach is time consuming and is feasible given a large data set with many missing values. *stop*
- **Use a global constant to fill in the missing value:** Replace all missing attribute values by the same constant such as a label. Example: "unknown", a new class?
- **Use the attribute mean to fill in the missing value:** Replace the missing values with mean or mode of non-missing values of that attribute or of same class.
- **Use the most probable value to fill in the missing value:** This may be determined with regression, inference-based such as Bayesian formula or decision tree.
- **Use the attribute mean or median for all samples belonging to the same class as the given tuple:** For example, if classifying customers according to credit risk, we may replace the missing value with the mean income value for customers in the same credit risk category as that of the given tuple.

6.4.2 Noisy Data

Noise: It is a random error or variance in a measured variable. Noisy data comes from the process of data collection, data entry, data transmission. In other words, incorrect attribute values may be due to-

- faulty data collection instruments
- data entry problems

HOW TO HANDLE THE NOISY DATA

BINNING METHOD

Binning method is used to smoothing data or to handle noisy data.

In this method, the data is first sorted and then the sorted values are distributed into a number of buckets or bins.

As binning methods consult the neighborhood of values, they perform local smoothing.

STEPS FOR BINNING METHOD

1. Sort the array of given data set.
2. Divides the range into N intervals, each containing the approximately same number of samples(Equal-depth partitioning).
3. Store mean/ median/ boundaries in each row.

There are three approaches to perform smoothing –

Smoothing by bin means : In smoothing by bin means, each value in a bin is replaced by the mean value of the bin.

Smoothing by bin median : In this method each bin value is replaced by its bin median value.

Smoothing by bin boundary : In smoothing by bin boundaries, the minimum and maximum values in a given bin are identified as the bin boundaries. Each bin value is then replaced by the closest boundary value.

Sorted data for price (in dollars): 4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29,

Smoothing by bin means:

- Bin 1: 9, 9, 9, 9
- Bin 2: 23, 23, 23, 23
- Bin 3: 29, 29, 29, 29

Smoothing by bin boundaries:

- Bin 1: 4, 4, 4, 15
- Bin 2: 21, 21, 25, 25
- Bin 3: 26, 26, 26, 34

Smoothing by bin median:

- Bin 1: 9, 9, 9, 9
- Bin 2: 24, 24, 24, 24
- Bin 3: 29, 29, 29, 29

6.4.4 Regression

Data can be smoothed by fitting the data into regression functions. *Linear regression* involves finding the “best” line to fit two attributes (or variables) so that one attribute can be used to predict the other. *Multiple linear regressions* is an extension of linear regression, where more than two attributes are involved and the data are fit into a multidimensional surface.

DATA INTEGRATION

Data Integration is a data preprocessing technique that involves combining data from multiple heterogeneous data sources into a coherent data store and provide a unified view of the data. These sources may include multiple data cubes, databases or flat files.

The data integration approach are formally defined as triple $\langle G, S, M \rangle$ where,

G stand for the global schema,

S stand for heterogeneous source of schema,

M stand for mapping between the queries of source and global schema.

There are mainly 2 major approaches for data integration – one is “tight coupling approach” and another is “loose coupling approach”.

Tight Coupling:

Here, a data warehouse is treated as an information retrieval component.

In this coupling, data is combined from different sources into a single physical location through the process of ETL – Extraction, Transformation and Loading.

Loose Coupling:

Here, an interface is provided that takes the query from the user, transforms it in a way the source database can understand and then sends the query directly to the source databases to obtain the result.

And the data only remains in the actual source databases.

Issues in Data Integration:

There are no of issues to consider during data integration: Schema Integration, Redundancy, Detection and resolution of data value conflicts. These are explained in brief as following below.

1. Schema Integration:

Schema integration integrates meta data from different sources. The semantic heterogeneity and structure of data pose great challenges in data integration. How can we match schema and objects from different sources? This is the essence of the entity identification problem,

Entity identification problem: identify real world entities from multiple data sources,

e.g. , A.cust-id=_B.cust-#

For example, How can the data analyst and computer be sure that customer id in one data base and customer number in another reference to the same attribute.

Redundant :

Redundant data occur often when integration of multiple databases is done. Another way to detect same attribute is

via., redundancy detection. An attribute is redundant if it can be derived from another attribute or a group of other attribute.

Inconsistencies in attribute can also cause redundancies in the resulting data set.

Some redundancies can be detected by correlation analysis.

Detection and resolution of data value conflicts:

A third important issue in data integration is detection and resolution . For example : for the same real world entity, attribute values from different sources are different.

Handling redundant data in data integration:

Redundant data occur often when integration of multiple databases

- The same attribute may have different names in different databases.
- One attribute may be “derived “ attribute in another table e.g. annual revenue

Redundant data may be able to be detected by correlational analysis.

Careful integration of the data from multiple sources may help reduce / avoid redundancies and inconsistencies and improve mining speed and quality.

6.7 Data Transformation

Data Transformation in terms of data mining is the process of changing the form or structure of existing attributes. Data Transformation is separate from data cleansing and data enrichment for data mining purpose because it does not correct existing data attributes data or add new attributes. Data transformation can involve the following:

Smoothing, which works to remove noise from the data. Techniques as regression, clustering, binning.

Attribute construction (or *feature construction*), where new attributes are constructed and added from the given set of attributes to help the mining process.

Aggregation, where summary or aggregation operations are applied to the data. For example, the daily sales data may be aggregated so as to compute monthly and annual total amounts. This step is typically used in constructing a data cube for data analysis at multiple abstraction levels.

Normalization, where the attribute data are scaled so as to fall within a smaller range, such as -1.0 to 1.0, or 0.0 to 1.0.

Discretization, where the raw values of a numeric attribute (e.g., *age*) are replaced by interval labels (e.g., 0–10, 11–20, etc.) or conceptual labels (e.g., *youth*, *adult*, *senior*).

Concept hierarchy generation for nominal data, where attributes such as *street* can be generalized to higher-level concepts, like *city* or *country*. Many hierarchies for nominal attributes are implicit within the database schema and can be automatically defined at the schema definition level.

6.7.1 Methods of Data Normalization

Data Transformation: Normalization

The various normalization methods are as follows:

- min-max normalization
- z-score normalization
- normalization by decimal scaling.

6.7.1.1 Min-max normalization

Min-max normalization performs a linear transformation on the original data. Suppose that min_A and max_A are the minimum and maximum values of an attribute, A . Min-max normalization maps a value, v_i , of A to v'_i in the range $[new_min_A, new_max_A]$ by computing

$$\text{Example } v'_i = \frac{v_i - min_A}{max_A - min_A} (new_max_A - new_min_A) + new_min_A$$

Suppose that the minimum and maximum values for the attribute monthly salary are Rs.20,000 and Rs.10,00,000, respectively. We would like to map monthly salary to the range. By min-max normalization, a value of Rs.63,500 for monthly salary is transformed to $83,500 - 20,000 / 10,00,000 - 20,000 (10. - 0) + 0 = 0.79375$.

6.7.1.2 z-score normalization

z-score normalization (or *zero-mean normalization*), the values for an attribute, A , are normalized based on the mean (i.e., average) and standard deviation of A . A value, v_i , of A is normalized to v'_i by computing

$$\text{Example } v'_i = \frac{v_i - mean_A}{stand_dev_A}$$

Suppose that the mean and standard deviation of the values of the attribute monthly salary are \$50,000 and \$18,000, respectively. With z-score transformation a value of Rs.83,500 monthly salary is transformed to $83,500 - 50,000 / 18,000 = 1.8611$.

6.7.1.3 Normalization by decimal scaling

Normalization by decimal scaling normalizes by moving the decimal point of values of attribute A . The number of decimal points moved depends on the maximum absolute value of A . A value, v_i , of A is normalized to v_i' by computing

Where j is the smallest integer such that $\text{Max}(|v_i'|) < 1$

$$v_i' = \frac{v_i}{10^j}$$

Example

Suppose that the recorded value of A range from -1045 to 1214. The maximum absolute value of A is 1214. To normalize by decimal scaling, we therefore divide each value by 1,000 so that -1045 normalizes to -0.1045 and 1214 normalizes to 0.1214.

DATA REDUCTION:

A data warehouse may store terra bytes of data: complex data analysis or mining may take a **very long time to run on the complete data set**.

Data reduction techniques obtain a reduced representation of the data set that is much **smaller in volume but yet produces the same(or almost the same) analytical results**.

Data reduction strategies:

Data reduction strategies include the following

1. **Data cube aggregation**: where aggregation operations are applied to the data in the construction of a data cube.
2. **Attribute subset selection**: where irrelevant, weak mechanisms are used to reduce the data size.
3. **Dimensionally reduction**: dimensionally reduction is the process of reducing the number of random variables or attributes under consideration
4. **Numerosity reduction**: replace the original data volume by alternative, smaller forms of data representation.
5. **Discretization and concept hierarchy generation**: where raw data values for attributes are replaced by range of higher conceptual levels.

Introduction to Data Cube

A Data cube as its name suggests is an extension of 2-Dimensional data cube or 2-dimensional matrix (column and rows) Whenever there are lots of complex data to be aggregated and there is a need to abstract the relevant or important data. There comes into picture the need for the data cube.

A Data cube is basically used to represent the specific information to be retrieved from a huge set of complex data. For example, You went to a shopping mall which has lots of items placed in different corners of the shopping mall and It is very difficult to find the item of need at the hour of need. Now if you are acknowledged about the order of the items placed in a shopping mall, buying of that item will become easy and hassle-free. This represents that a data cube with

perfect dimensions and higher value ranges or we can say a reference to three-dimensional data as well.

What is Data Cube?

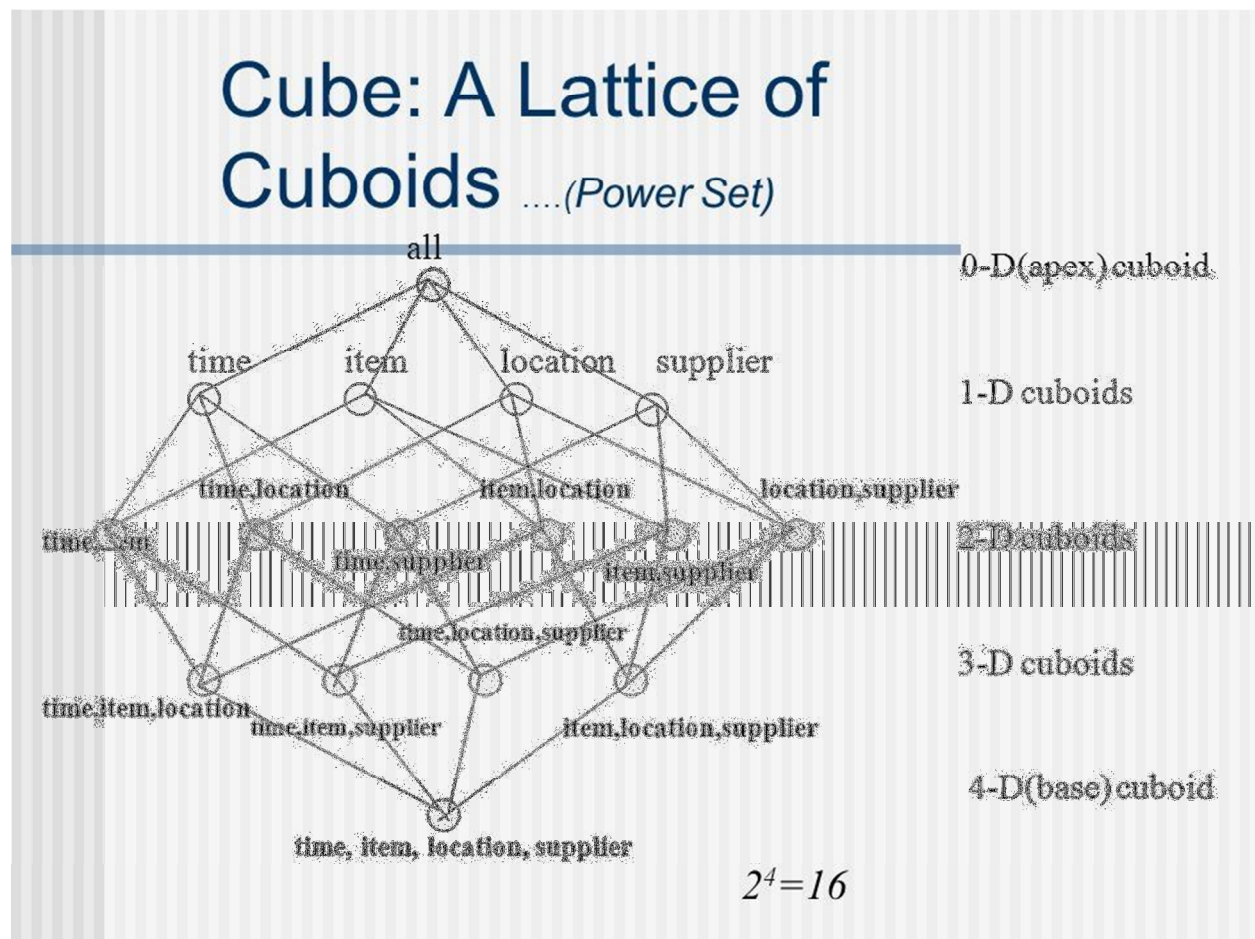
It has many characteristics are as follows:

- It can go very far beyond to include many more dimensions.
- Improvises business strategies by analysis of all the data.
- It helps to get the latest market scenario by establishing trends and performance analysis.
- It plays a very pivotal role by creating intermediate data cubes to serve the requirements and to bridge the gap between the data warehouse and all the reporting tool, particularly in a data warehouse reporting tool.
- In other phases there will be source input which is simultaneously monitored and administered, the goal is to create a connection and end to end flow between source to destination with intermediate data cubes interacting with servers.

This technique is used to aggregate data in a simpler form. For example, imagine that information you gathered for your analysis for the years 2012 to 2014, that data includes the revenue of your company every three months. They involve you in the annual sales, rather than the quarterly average, So we can summarize the data in such a way that the resulting data summarizes the total sales per year instead of per quarter. It summarizes the data.

Data cube aggregation

Data cubes store multidimensional aggregated information. Data cube aggregation, where aggregation operations are applied to the data in the construction of a data cube. The cube can be created in three ways:



1. Base cuboids: the cube created at the lowest level of abstraction
2. Lattice of cuboids: data cubes created for varying levels of abstraction are often referred to as lattice cuboids
3. Apex cuboids: a cube at the highest level of abstraction is the apex cuboid

This technique is used to aggregate data in a simpler form. For example, imagine that information you gathered for your analysis for the years 2012 to 2014, that data includes the revenue of your company every three months. They involve you in the annual sales, rather than the quarterly average, So we can summarize the data in such a way that the resulting data summarizes the total sales per year instead of per quarter. It summarizes the data.

Attribute Subset Selection in Data Mining

Attribute subset Selection is a technique which is used for data reduction in data mining process. Data reduction reduces the size of data so that it can be used for analysis purposes more efficiently.

Need of Attribute Subset Selection-

The data set may have a large number of attributes. But some of those attributes can be irrelevant or redundant. The goal of attribute subset selection is to find a minimum set of attributes such that dropping of those irrelevant attributes does not much affect the utility of data and the cost of data analysis could be reduced. Mining on a reduced data set also makes the discovered pattern easier to understand.

1. Step-wise Forward Selection –

The procedure starts with an empty set of attributes as the reduced set. The best of the original attributes is determined and added to the reduced set. At each subsequent iteration or step, the best of the remaining original attributes is added to the set.

Suppose there are the following attributes in the data set in which few attributes are redundant.

Initial attribute Set: {X1, X2, X3, X4, X5, X6}

Initial reduced attribute set: {}

Step-1: {X1}

Step-2: {X1, X2}

Step-3: {X1, X2, X5}

Final reduced attribute set: {X1, X2, X5}

2. Step-wise Backward elimination method –

The procedure start with the full set of attributes. At each step, it removes the worst attribute remaining in the set.

Suppose there are the following attributes in the data set in which few attributes are redundant.

Initial attribute Set: {X1, X2, X3, X4, X5, X6}

Initial reduced attribute set: {X1, X2, X3, X4, X5, X6 }

Step-1: {X1, X2, X3, X4, X5}

Step-2: {X1, X2, X3, X5}

Step-3: {X1, X2, X5}

Final reduced attribute set: {X1, X2, X5}

3. Combination of forwarding and Backward elimination –

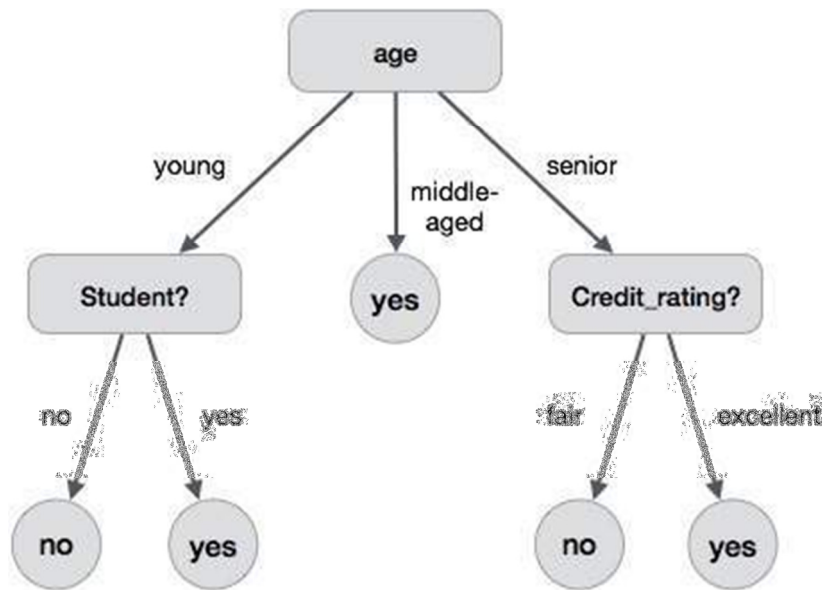
The stepwise forward selection and backward elimination methods can be combined so that , at each step, the procedure selects the best attribute and removes the worst from among the remaining attributes

saving time and making the process faster.

4. Decision Tree Induction:

A decision tree is a structure that includes a root node, branches, and leaf nodes. Each internal node denotes a test on an attribute, each branch denotes the outcome of a test, and each leaf node holds a class label. The topmost node in the tree is the root node.

The following decision tree is for the concept buy_computer that indicates whether a customer at a company is likely to buy a computer or not. Each internal node represents a test on an attribute. Each leaf node represents a class.



1. Data Compression:

The data compression technique reduces the size of the files using different encoding mechanisms (Huffman Encoding & run-length Encoding). We can divide it into two types based on their compression techniques.

- Lossless Compression –
Encoding techniques (Run Length Encoding) allows a simple and minimal data size reduction. Lossless data compression uses algorithms to restore the precise original data from the compressed data.
- Lossy Compression –
Methods such as Discrete Wavelet transform technique, PCA (principal component analysis) are examples of this compression. For e.g., JPEG image format is a lossy compression, but we can find the meaning equivalent to the original the image. In lossy-data

compression, the decompressed data may differ to the original data but are useful enough to retrieve information from them.

2. Numerosity Reduction:

In this reduction technique the actual data is replaced with mathematical models or smaller representation of the data instead of actual data, it is important to only store the model parameter. Or non-parametric method such as clustering, histogram, sampling. For More Information on Numerosity Reduction Visit the link below:

3. Discretization & Concept Hierarchy Operation:

Techniques of data discretization are used to divide the attributes of the continuous nature into data with intervals. We replace many constant values of the attributes by labels of small intervals. This means that mining results are shown in a concise, and easily understandable way.

- Top-down discretization –

If you first consider one or a couple of points (so-called breakpoints or split points) to divide the whole set of attributes and repeat of this method up to the end, then the process is known as top-down discretization also known as splitting.

- Bottom-up discretization –

If you first consider all the constant values as split-points, some are discarded through a combination of the neighbourhood values in the interval, that process is called bottom-up discretization.

Concept Hierarchies:

It reduces the data size by collecting and then replacing the low-level concepts (such as 43 for age) to high-level concepts (categorical variables such as middle age or Senior).

For numeric data following techniques can be followed:

- **Binning** –

Binning is the process of changing numerical variables into categorical counterparts. The number of categorical counterparts depends on the number of bins specified by the user.

- Histogram analysis –

Like the process of binning, the histogram is used to partition the value for the attribute X, into disjoint ranges called brackets. There are several partitioning rules:

1. Equal Frequency partitioning: Partitioning the values based on their number of occurrences in the data set.
2. Equal Width Partitioning: Partitioning the values in a fixed gap based on the number of bins i.e. a set of values ranging from 0-20.
3. Clustering: Grouping the similar data together.

CLASSIFICATION:

Classification is a technique where we categorize data into a given number of classes.

The main goal of a classification problem is to identify the category/class to which a new data will fall under.

Classification analysis is the organization of data in given classes. Also known as **supervised** classification, the classification uses given class labels to order the objects in the data collection

Why Classification?

Very large databases are becoming the norm in today's world of big data. Imagine a database with terabytes of data—a terabyte is one trillion bytes of data. Facebook alone crunches 600 terabytes of new data every single day (as of 2014, the last time it reported these specs). The primary challenge of big data is how to make sense of it.

And sheer volume is not the only problem: big data also tends to be diverse, unstructured and fast-changing. Consider audio and video data, social media posts, 3D data, or geospatial data. This kind of data is not easily categorized or organized.

To meet this challenge, a range of automatic methods for extracting useful information has been developed, among them classification.

Classification approaches normally use a training set where all objects are already associated with known class labels.

The classification algorithms learns from the training set and builds a model. The model is used to classify new objects.

For example, after starting a credit policy, the video store managers could analyze the customers behaviors vis-à-vis their credit, and labels accordingly the customers who received credits with three possible labels “safe”, “risky”, “very risky” . the classification

analysis would generate a model that could be used to either accept or reject credit requests in the future.

How Classification Works

An analyst's goal is to create a set of classification rules that answer a question, make a decision, or predict behavior. To start, a set of training data is developed that contains a certain set of attributes as well as the likely outcome. The job of the classification algorithm is to discover how that set of attributes reaches its conclusion.

Consider a credit-card company trying to determine which prospects should receive a credit card offer.

The company's training data might include:

Name	Age	Gender	Annual Income	Credit Card Offer
John Doe	25	M	\$39,500	No
Jane Doe	56	F	\$125,000	Yes

Training Data

The predictor columns Age, Gender, and Annual Income determine the value of the "predictor attribute" Credit Card Offer. In a training set, the predictor attribute is known. The classification algorithm then tries to determine how the value of the predictor attribute was reached: what relationships exist between the predictors and the decision? It will develop a set of prediction rules, usually an IF/THEN statement.

this is a simple example, and the algorithm would need a far larger data sampling than the two records shown here. Further, the prediction rules are likely to be far more complex, including sub-rules to capture attribute details.

Next, the algorithm is given a "prediction set" of data to analyze, but this set lacks the prediction attribute (or decision):

Name	Age	Gender	Annual Income	Credit Card Offer
Jack Frost	42	M	\$88,000	? (Y)
Mary Murray	16	F	\$0	? (N)

Example

A classification task begins with a data set in which the class assignments are known.

For example, a classification model that predicts credit risk could be developed based on observed data for many loan applicants over a period of time.

In addition to the historical credit rating, the data might track employment history, home ownership or rental, years of residence, number and type of investments, and so on. Credit rating would be the target, the other attributes would be the predictors, and the data for each customer would constitute a case.

classifier: An algorithm that maps the input data to a specific category.

Classification model: A classification model tries to draw some conclusion from the input values given for training. It will predict the class labels/categories for the new data.

Feature: A feature is an individual measurable property of a phenomenon being observed.

Binary Classification: Classification task with two possible outcomes. Eg: Gender classification (Male / Female)

Multi-class classification: Classification with more than two classes. In multi class classification each sample is assigned to one and only one target label. Eg: An animal can be cat or dog but not both at the same time

Multi-label classification: Classification task where each sample is mapped to a set of target labels (more than one class). Eg: A news article can be about sports, a person, and location at the same time.

Classification—A Two-Step Process

1. **Model construction:** describing a set of predetermined classes

Each tuple/sample is assumed to belong to a predefined class, as determined by the class label attribute

The set of tuples used for model construction is training set

The model is represented as classification rules, decision trees, or mathematical formulae

2. **Model usage:** for classifying future or unknown objects

Estimate accuracy of the model

The known label of test sample is compared with the classified result from the model

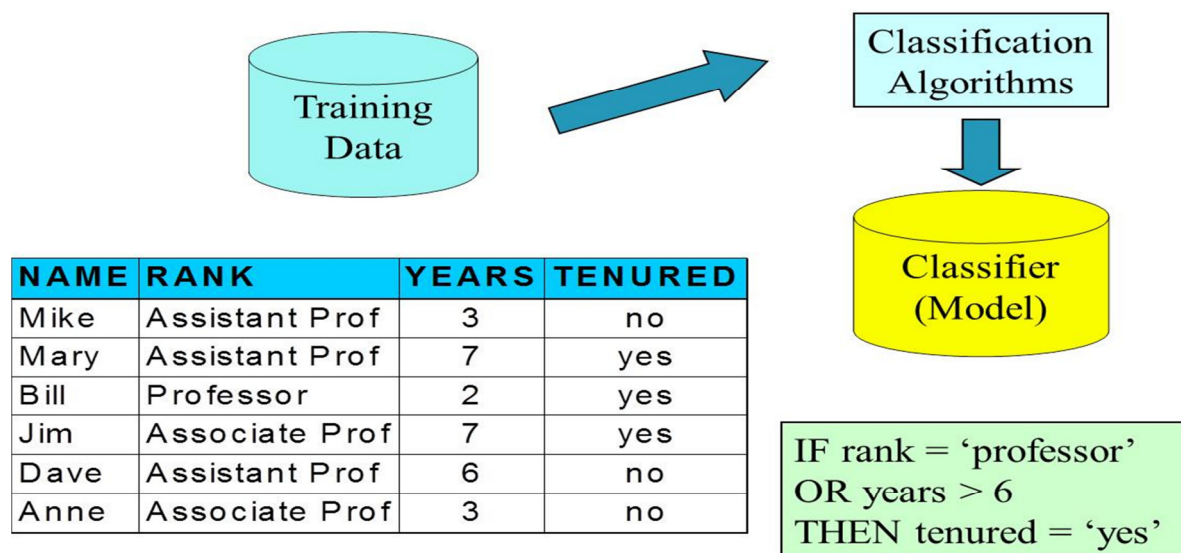
Accuracy rate is the percentage of test set samples that are correctly classified by the model

Test set is independent of training set (otherwise overfitting)

If the accuracy is acceptable, use the model to classify new data

Note: If *the test set* is used to select models, it is called validation (test) set

Process (1): Model Construction



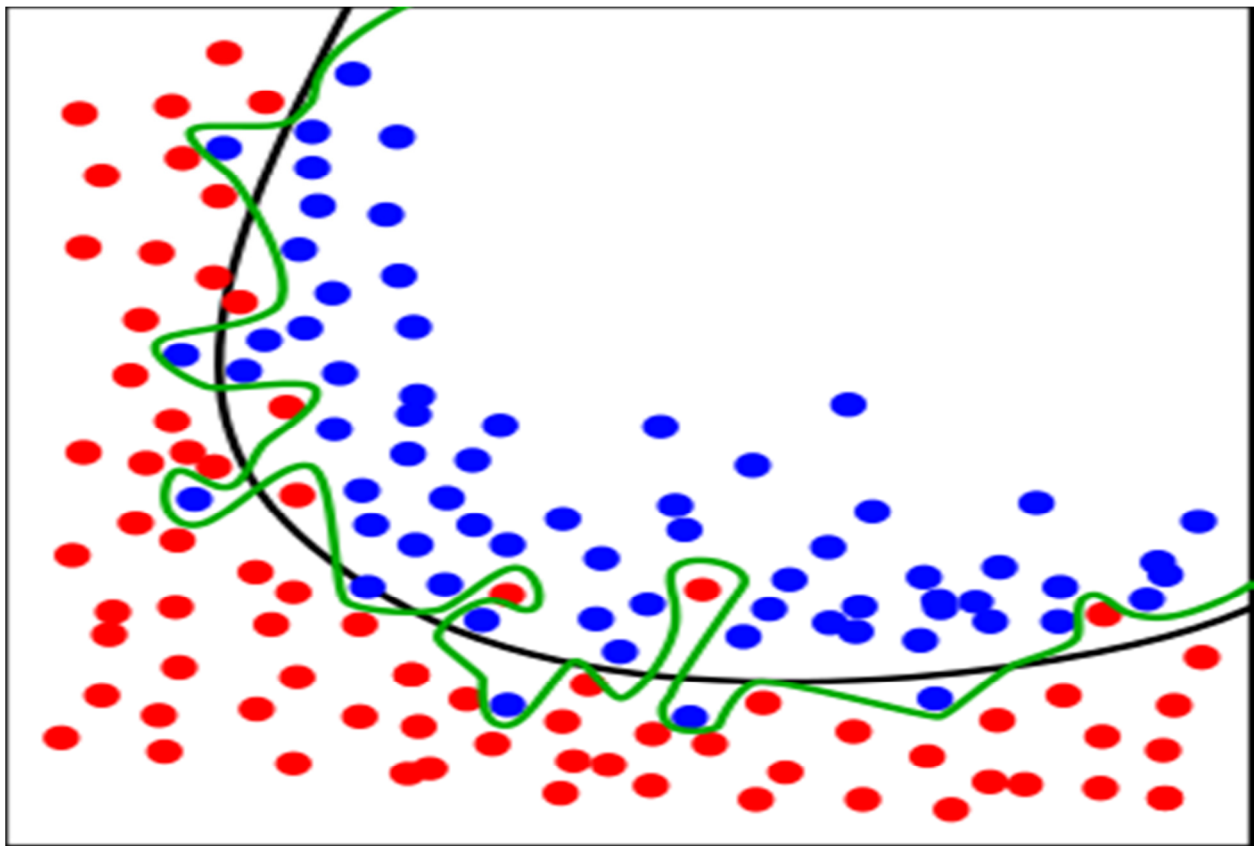
Accuracy is one metric for evaluating classification models. Informally, accuracy is the fraction of predictions our model got right. Formally, accuracy has the following definition:

$$\text{Accuracy} = \frac{\text{Number of correct prediction}}{\text{Total number of prediction}}$$

ISSUES REGARDING CLASSIFICATION

- The following preprocessing steps may be applied to the data which helps to improve the accuracy, efficiency and scalability of the classification
- **Data cleaning:**
 - This refers to the processes data in order to reduce noise and handling missing values.
 - Relevance Analysis(feature selection)
 - remove the irrelevant or redundant attributes
 - Data Transformation and reduction
 - Generalize and / or normalize data
- **Overfitting:** Good performance on the training data, poor generalization to other data.
- **Underfitting:** Poor performance on the training data and poor generalization to other data
- Examples of Overfitting

- Let's say we want to predict if a student will land a job interview based on her resume.
- Now, assume we train a model from a dataset of 10,000 resumes and their outcomes.
- Next, we try the model out on the original dataset, and it predicts outcomes with 99% accuracy... wow!
- But now comes the bad news.
- When we run the model on a new ("unseen") dataset of resumes, we only get 50% accuracy... uh-oh!



Techniques Reduce Overfitting

1. Increase the Training Data
2. Reduce the model Complexity

Techniques reduce underfitting

1. Increase model complexity
2. Increase the number of features
3. Remove the noise from the data

NAIVE BAYES ALGORITHM

Bayes theorem provides a way of calculating posterior probability $P(c|x)$ from $P(c)$, $P(x)$ and $P(x|c)$. Look at the equation below:

Diagram illustrating Bayes' Theorem:

$$P(c | x) = \frac{P(x | c) P(c)}{P(x)}$$

Labels and arrows:

- Likelihood** points to $P(x | c)$.
- Class Prior Probability** points to $P(c)$.
- Posterior Probability** points to $P(c | x)$.
- Predictor Prior Probability** points to $P(x)$.

$$P(c | \mathbf{X}) = P(x_1 | c) \times P(x_2 | c) \times \cdots \times P(x_n | c) \times P(c)$$

$P(c|x)$ is the posterior probability of class (c, target) given predictor (x, attributes).

$P(c)$ is the prior probability of class.

$P(x | c)$ is the likelihood which is the probability of predictor given class.

$P(x)$ is the prior probability of predictor.

EXAMPLE-2

CONDITION:

2M<= HEIGHT TALL

1.7M<HEIGHT<2M MEDIUM

HEIGHT<=1.7M SHORT

Table 1:

S.NO	NAME	GENDER	HEIGHT	OUTPUT
1	KRISTINA	F	1.6 M	SHORT
2	JIM	M	2M	TALL
3	MAGGIE	F	1.9M	MEDIUM
4	MARTHA	F	1.88.M	MEDIUM
5	STEPHANIE	F	1.7M	SHORT
6	BOB	M	1.85M	MEDIUM
7	KATHY	F	1.6M	SHORT

8	DAVE	M	1.7M	SHORT
9	WORTH	M	2.2M	TALL
10	STEVEN	M	2.1M	TALL
11	DEBBIE	F	1.8M	MEDIUM
12	TODD	M	1.95M	MEDIUM
13	KIM	M	1.9M	MEDIUM
14	AMY	F	1.8.M	MEDIUM
15	WYNETTE	F	1.75M	MEDIUM

Step 1: Convert the data set into a frequency table

Step 2: Create Likelihood table by finding the probabilities like red

Step 3: Now, use Naive Bayesian equation to calculate the posterior probability

for each class. The class with the highest posterior probability is the outcome of prediction.

Using the output of the classification results of the table

Short	-	4	Male: 1	- Female:3
Medium	-	8	Male:2	- Female:6
Tall	-	3	Male: 3	- Female:0

$$P(\text{SHORT}) = 4 / 15 = 0.267$$

$$P(\text{MEDIUM}) = 8 / 15 = 0.533$$

$$P(\text{TALL}) = 3 / 15 = 0.2$$

We divide the height attribute into six ranges

[0-1.6],[1.6-1.7],[1.7-1.8],[1.8-1.9],[1.9-2.0], [2.0- above]

Table 2: shows the counts and subsequent probabilities associated with the attributes. With these training data, we estimate the prior probabilities.

$$P(\text{SHORT}) = 4/15 = 0.267$$

$$P(\text{MEDIUM}) = 8/15 = 0.533$$

$$P(\text{TALL}) = 3/15 = 0.2$$

ATTRIBUTE	VALUE	COUNT			PROBABILITY		
		Short	Medium	Tall	Short	Medium	Tall
Gender	M	1	2	3	$\frac{1}{4}$	$\frac{2}{8}$	$\frac{3}{8}$
	F	3	6	0	$\frac{3}{4}$	$\frac{6}{8}$	$\frac{0}{8}$
Height	[0-1.6]	2	0	0	$\frac{2}{4}$	0	0
	]1.6-1.7]	2	0	0	$\frac{2}{4}$	0	0
	[1.7-1.8]	0	3	0	0	$\frac{3}{8}$	0
	[1.8-1.9]	0	4	0	0	$\frac{4}{8}$	0
	[1.9-2.0]	0	1	1	0	$\frac{1}{8}$	$\frac{1}{8}$
	[2.0-above]	0	0	2	0	0	$\frac{2}{8}$

We use these values to classify a new tuple.

Example we wish to **classify** $t = \{\text{adam}, \text{M}, 1.95\text{m}\}$. by using these values and the associated probabilities of gender height, we obtain the following estimates

$$P(t | \text{short}) = \frac{1}{4} \times 0 = 0$$

$$P(t | \text{medium}) = \frac{2}{8} \times \frac{1}{8} = 0.031$$

$$P(t | \text{tall}) = \frac{3}{8} \times \frac{1}{8} = 0.333$$

Combining these we get,

$$\text{Likelihood of being short} = 0 \times 0.267 = 0$$

$$\text{Likelihood of being medium} = 0.031 \times 0.5333 = 0.0166$$

$$\text{Likelihood of being tall} = 0.33 \times 0.2 = 0.066$$

We estimate $P(t)$ summing up these individual likelihood values since t will be either short or medium or tall

$$P(t) = 0 + 0.0166 + 0.066 = 0.0826$$

Finally, we obtain the actual probabilities of each event

$$P(\text{short} | t) = \frac{0 \times 0.0267}{0.0826} = 0$$

$$P(\text{medium} | t) = \frac{0.031 \times 0.533}{0.0826} = 0.2$$

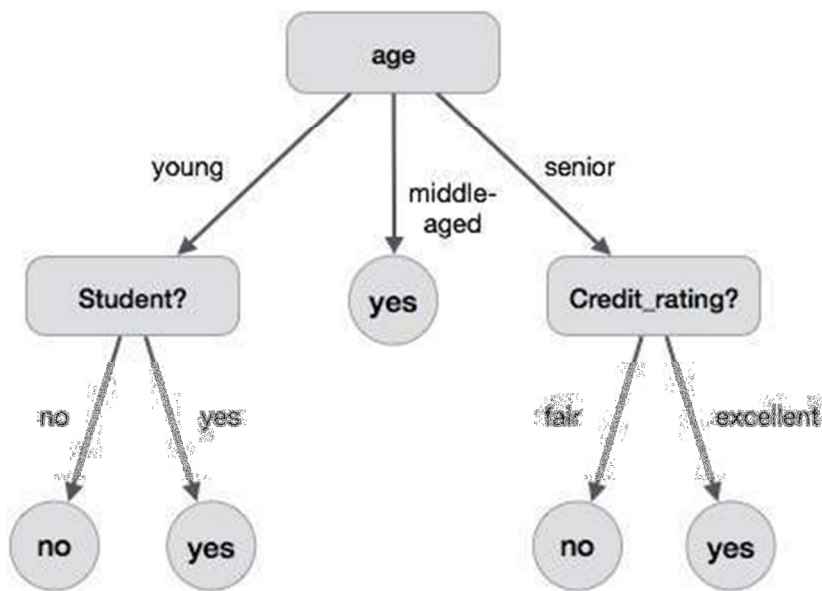
$$P(\text{tall} | t) = \frac{0.333 \times 0.2}{0.0826} = 0.799$$

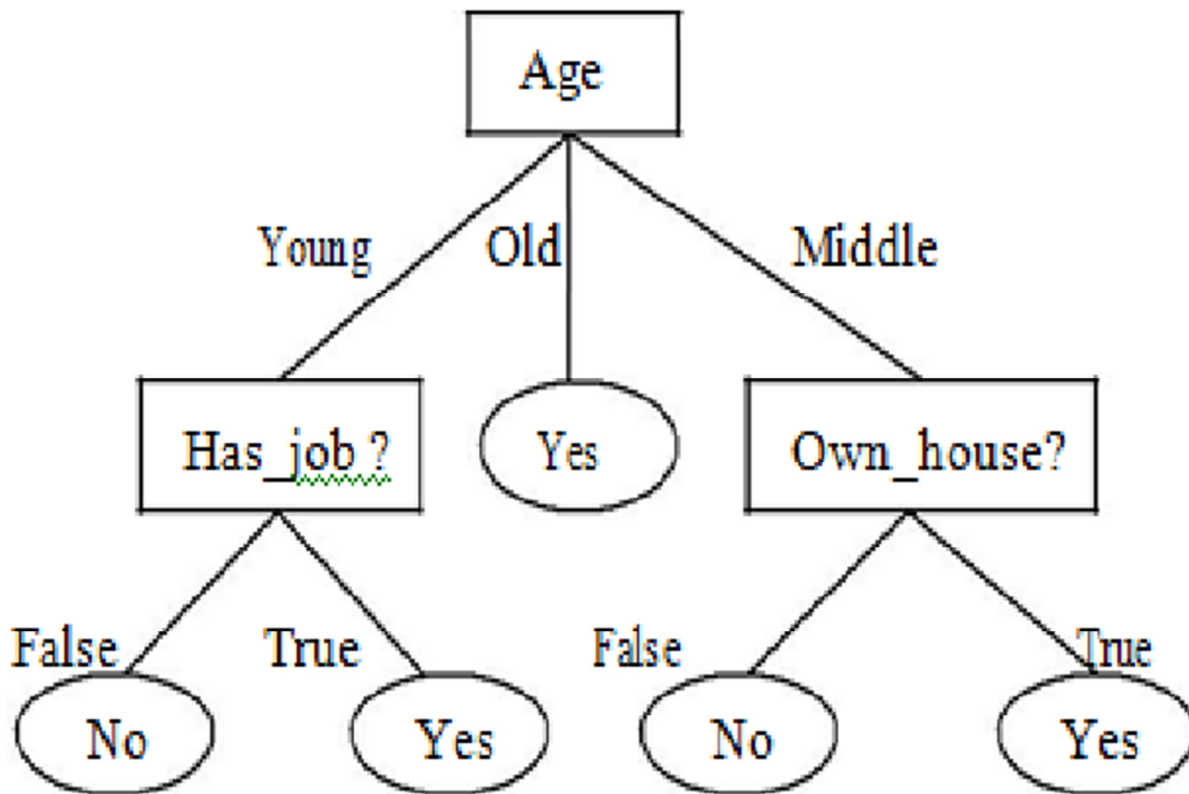
Therefore based on the probabilities, we classify the new tuple as **Tall** because it has the highest probability.

Decision Tress

A decision tree is a structure that includes a root node, branches, and leaf nodes. Each internal node denotes a test on an attribute, each branch denotes the outcome of a test, and each leaf node holds a class label. The topmost node in the tree is the root node.

The following decision tree is for the concept `buy_computer` that indicates whether a customer at a company is likely to buy a computer or not. Each internal node represents a test on an attribute. Each leaf node represents a class.





The benefits of having a decision tree are as follows –

- It does not require any domain knowledge.
- It is easy to comprehend.
- The learning and classification steps of a decision tree are simple and fast.

Decision Trees are a type of Supervised Machine Learning

where the data is continuously split according to a certain parameter.

The tree can be explained by two entities, namely decision nodes and leaves. The leaves are the decisions or the final outcomes. And the decision nodes are where the data is split. (Fitness example)

There are two main types of Decision Trees:

Classification trees (Yes/No types)

Regression trees (Continuous data types)

There are many algorithms out there which construct Decision Trees, but one of the best is called as ID3 Algorithm.

ID3 Stands for Iterative(repeat) Dichotomiser(divide) 3.

Entropy

Entropy is the measures of impurity, disorder or uncertainty, messy of data

Entropy controls how a Decision Tree decides to split the data. It actually effects how a Decision Tree draws its boundaries.

Entropy, also called as Shannon Entropy is denoted by $H(S)$ for a finite set S , is the measure of the amount of uncertainty or randomness in data.

$$H(S) = \sum_{x \in X} p(x) \log_2 \frac{1}{p(x)}$$

Intuitively, it tells us about the predictability of a certain event. Example, consider a coin toss whose probability of heads is 0.5 and probability of tails is 0.5. Here the entropy is the highest possible, since there's no way of determining what the outcome might be. Alternatively, consider a coin which has heads on both the sides, the entropy of such an event can be predicted perfectly since we know beforehand that it'll always be heads. In other words, this event has no randomness hence it's entropy is zero.

In particular, lower values imply less uncertainty while higher values imply high uncertainty.

Information Gain

Information gain (IG) measures how much "information" a feature gives us about the class. Information gain is the main key that is used by Decision Tree Algorithms to construct a Decision Tree. Decision Trees algorithm will always tries to maximize Information gain. An attribute with highest Information gain will tested/split first.

Information gain is also called as Kullback-Leibler divergence denoted by $IG(S,A)$ for a set S is the effective change in entropy after deciding on a particular attribute A . It measures the relative change in entropy with respect to the independent variables.

$$IG(S,A) = H(S) - H(S,A)$$

Alternatively,

$$IG(S,A) = H(S) - \sum_{i=0}^n P(x) * H(x)$$

where $IG(S, A)$ is the information gain by applying feature A . $H(S)$ is the Entropy of the entire set, while the second term calculates the Entropy after applying the feature A , where $P(x)$ is the probability of event x .

Let's understand this with the help of an example

Consider a piece of data collected over the course of 14 days where the features are Outlook, Temperature, Humidity, Wind and the outcome variable is whether Cricket was played on the day. Now, our job is to build a predictive model which takes in above 4 parameters and predicts whether Cricket will be played on the day. We'll build a decision tree to do that using ID3 algorithm.

Day	Outlook	Temperature	Humidity	Wind	Play
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes

D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

ID3 Algorithm will perform following tasks recursively

1. Create root node for the tree
2. If all examples are positive, return leaf node 'positive'
3. Else if all examples are negative, return leaf node 'negative'
4. Calculate the entropy of current state $H(S)$
5. For each attribute, calculate the entropy with respect to the attribute 'x' denoted by $H(S, x)$
6. Select the attribute which has maximum value of $IG(S, x)$
7. Remove the attribute that offers highest IG from the set of attributes
8. Repeat until we run out of all attributes, or the decision tree has all leaf nodes.

Now we'll go ahead and grow the decision tree. The initial step is to calculate $H(S)$, the Entropy of the current state. In the above example, we can see in total there are 5 No's and 9 Yes's.

Yes	No	Total
9	5	14

$$\text{Entropy}(S) = \sum -p(I) \cdot \log_2 p(I)$$

$$\text{Entropy}(\text{Decision}) = -p(\text{Yes}) \cdot \log_2 p(\text{Yes}) - p(\text{No}) \cdot \log_2 p(\text{No})$$

$$\text{Entropy}(\text{Decision}) = -(9/14) \cdot \log_2(9/14) - (5/14) \cdot \log_2(5/14) = 0.940$$

Remember that the Entropy is 0 if all members belong to the same class, and 1 when half of them belong to one class and other half belong to other class that is perfect randomness. Here it's 0.94 which means the distribution is fairly random.

Now the next step is to choose the attribute that gives us highest possible Information Gain which we'll choose as the root node.

Let's start with 'Wind'

$$IG(S, \text{Wind}) = H(S) - \sum_{i=0}^n P(x) * H(x)$$

where 'x' are the possible values for an attribute. Here, attribute 'Wind' takes two possible values in the sample data, hence $x = \{\text{Weak}, \text{Strong}\}$

We'll have to calculate:

Amongst all the 14 examples we have 8 places where the wind is weak and 6 where the wind is Strong.

Wind = Weak	Wind = Strong	Total
8	6	14

$$\begin{aligned}P(S_{\text{weak}}) &= \frac{\text{Number of Weak}}{\text{Total}} \\&= \frac{8}{14} \\P(S_{\text{strong}}) &= \frac{\text{Number of Strong}}{\text{Total}} \\&= \frac{6}{14}\end{aligned}$$

Now out of the 8 Weak examples, 6 of them were 'Yes' for Play cricket and 2 of them were 'No' for 'Play Cricket'. So, we have,

$$\begin{aligned}\text{Entropy}(S_{\text{weak}}) &= -\left(\frac{6}{8}\right)\log_2\left(\frac{6}{8}\right) - \left(\frac{2}{8}\right)\log_2\left(\frac{2}{8}\right) \\&= 0.811\end{aligned}$$

Similarly, out of 6 Strong examples, we have 3 examples where the outcome was 'Yes' for Play Cricket and 3 where we had 'No' for Play Cricket.

$$\begin{aligned}\text{Entropy}(S_{\text{strong}}) &= -\left(\frac{3}{6}\right)\log_2\left(\frac{3}{6}\right) - \left(\frac{3}{6}\right)\log_2\left(\frac{3}{6}\right) \\&= 1.000\end{aligned}$$

Remember, here half items belong to one class while other half belong to other. Hence we have perfect randomness.

Now we have all the pieces required to calculate the Information Gain,

$$\begin{aligned}IG(S, Wind) &= H(S) - \sum_{i=0}^n P(x) * H(x) \\IG(S, Wind) &= H(S) - P(S_{weak}) * H(S_{weak}) - P(S_{strong}) * H(S_{strong}) \\&= 0.940 - \left(\frac{8}{14}\right)(0.811) - \left(\frac{6}{14}\right)(1.00) \\&= 0.048\end{aligned}$$

Which tells us the Information Gain by considering 'Wind' as the feature and give us information gain of 0.048. Now we must similarly calculate the Information Gain for all the features.

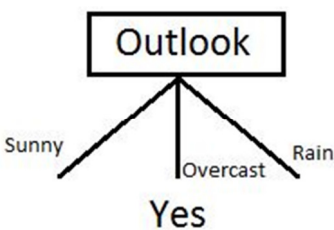
$$IG(S, Outlook) = 0.246$$

$$IG(S, Temperature) = 0.029$$

$$IG(S, Humidity) = 0.151$$

$$IG(S, Wind) = 0.048 \text{ (Previous example)}$$

We can clearly see that $IG(S, Outlook)$ has the highest information gain of 0.246, hence we chose Outlook attribute as the root node. At this point, the decision tree looks like.



Here we observe that whenever the outlook is Overcast, Play Cricket is always 'Yes', it's no coincidence by any chance, the simple tree resulted because of the highest information gain is given by the attribute Outlook.

Now how do we proceed from this point? We can simply apply recursion, you might want to look at the algorithm steps described earlier.

Now that we've used Outlook, we've got three of them remaining Humidity, Temperature, and Wind. And, we had three possible values of Outlook: Sunny, Overcast, Rain. Where the Overcast node already ended up having leaf node 'Yes', so we're left with two subtrees to compute: Sunny and Rain.

Next step would be computing $H(S_{\text{sunny}})$.

Table where the value of Outlook is Sunny looks like:

Temperature	Humidity	Wind	Play Cricket
Hot	High	Weak	No
Hot	High	Strong	No
Mild	High	Weak	No
Cool	Normal	Weak	Yes
Mild	Normal	Strong	Yes

$$H(S_{\text{sunny}}) = \left(\frac{3}{5}\right) \log_2 \left(\frac{3}{5}\right) - \left(\frac{2}{5}\right) \log_2 \left(\frac{2}{5}\right) = 0.96$$

In the similar fashion, we compute the following values

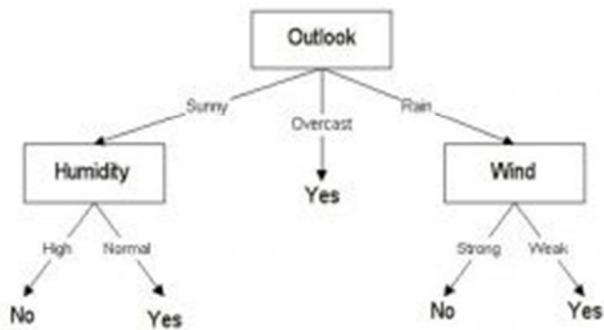
$$IG(S_{\text{sunny}}, \text{Humidity}) = 0.96$$

$$IG(S_{\text{sunny}}, \text{Temperature}) = 0.57$$

$$IG(S_{\text{sunny}}, \text{Wind}) = 0.019$$

As we can see the highest Information Gain is given by Humidity. Proceeding in the same way with S_{rain} will give us Wind as the one with highest information gain. The final Decision Tree looks something like this.

The final Decision Tree looks something like this.



Conclusion:

1. Entropy to measure discriminatory power of an attribute for classification task. It defines the amount of randomness in attribute for classification task. Entropy is minimal means the attribute appears close to one class and have a good discriminatory power for classification
2. Information Gain to rank attribute for filtering at given node in the tree. The ranking is based on high information gain entropy in decreasing order.
3. The recursive ID3 algorithm that creates a decision tree.